Minimization of Objective function to obtain BEST model parameters

$$J(W) = \frac{1}{2}(Y - \hat{Y})^T(Y - \hat{Y}) + \frac{1}{2}\lambda W^T W = \frac{1}{2}(Y - XW)^T(Y - XW) + \frac{1}{2}\lambda W^T W$$

$$J(W) = \frac{1}{2}[Y^T Y + (XW)^T(XW) - Y^T(XW) - (XW)^T Y] + \frac{1}{2}\lambda W^T W$$

$$J(W) = \frac{1}{2}[Y^T Y + (XW)^T(XW) - 2(XW)^T Y] + \frac{1}{2}\lambda W^T W$$

$\frac{\partial J(W)}{\partial W} = 0$ Get Best W that minimizes the Objective Function J(W)

Remember

$$\frac{\partial W^T W}{\partial W} = 2W$$

$$\frac{\partial J(W)}{\partial W} = 0 = \frac{1}{2}[(XW)^T(X) + (X)^T(XW) - 2(X)^T Y] + \lambda W$$

$$\frac{\partial J(W)}{\partial W} = 0 = \frac{1}{2}[2X^T XW - 2X^T Y] + \lambda IW$$

$$(X^T X + \lambda I)W = X^T Y \Rightarrow \underline{W} = (X^T X + \lambda I)^{-1} X^T Y$$